

Simplifying Surds

AQA · KS4 Year 10–11

Name: _____ Class: _____ Date: _____

Simplify surds by finding the largest square factor. Express surds in their simplest form $a\sqrt{b}$. Understand why surds are 'exact' values and when to leave answers in surd form.

VISUAL EXAMPLE

$$\begin{aligned}72 &= (36 \times 2) \\&= 36 \times 2 \\&= 6\sqrt{2}\end{aligned}$$

Simplifying a surd: split off the largest square factor.

COMMON MISTAKES

Not finding the LARGEST square factor — $72 = (36 \times 2) = 6\sqrt{2}$, not $(4 \times 18) = 2\sqrt{18}$ (not fully simplified).

Thinking $(a + b) = a + b$ — this is WRONG. $(9 + 16) \neq 9 + 16$.

Writing $4 \times 3 = 12$ is correct, but the point of simplifying is to go the OTHER way: $12 = 4 \times 3 = 2\sqrt{3}$.

Forgetting that $a \times a = a^2$ (not a^2).

METHOD

- 1 Find the largest perfect square that is a factor of the number under the $\sqrt{\quad}$.
- 2 Split: $n = (\text{square factor} \times \text{remainder})$.
- 3 Use $(a \times b) = a \times b$.
- 4 Take the square root of the perfect square factor out.
- 5 Write answer as $k\sqrt{m}$ where m has no square factors.

WORKED EXAMPLE

Simplify $\sqrt{72}$.

Step 1 — Largest square factor of 72: 36 (since $36 \times 2 = 72$).

Step 2 — $\sqrt{72} = \sqrt{(36 \times 2)}$

Step 3 — $= \sqrt{36} \times \sqrt{2}$

Step 4 — $= 6\sqrt{2}$

Check: $6^2 \times 2 = 36 \times 2 = 72$

- a) 12
- b) 18
- c) 20
- d) 50

HOW TO START

Find the largest square number that divides into the number under the root (e.g. 4, 9, 16, 25, 36...).

Split it: $12 = (4 \times 3) = 4 \times 3$.

Take the square root of the square factor: $4 = 2$, so $12 = 2 \times 3$.

ANSWERS

- (a) =
- (b) =
- (c) =
- (d) =

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = a \times b$ | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

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Simplify:

(2 marks)

- a) 8 b) 32 c) 45 d) 75

HOW TO START

Same method: largest square factor first.

For 8, the largest square factor is 4: $8 = 4 \times 2 = 2^2 \cdot 2$.

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k \sqrt{m}$ (m has no square factors)

3 Simplify:

(3 marks)

- a) 98 b) 200 c) 128 d) 300

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

4 Simplify:

(3 marks)

- a) (4×7) b) $3 \ 12$ c) $2 \ 50$ d) $5 \ 8$

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____
- (d) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

5 Simplify fully:

(3 marks)

a) $48 + 12$ b) $75 - 27$ c) $2 \sqrt{18} + 3 \sqrt{2}$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k \sqrt{m}$ (m has no square factors)

6 Show that:

(4 marks)

a) $80 = 4 \times 5$ b) $3 \times 28 = 6 \times 7$ c) $180 = 6 \times 5$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k \times m$ (m has no square factors)

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(4 marks)

- a) Find the exact side length. (1 mark)

- b) Find the exact perimeter. (1 mark)

- c) A second square has side length 3.2 cm. Which square is larger? Show working. (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = a \times b$ | 4 Square-root the square factor | 5 Answer: $k \sqrt{m}$ (m has no square factors)

- 8** a) Show that $(a^2b) = a \sqrt{b}$ for any positive a and b . (1 mark) (5 marks)
- b) Simplify $(50x^4y^2)$ where $x, y > 0$. (2 marks)
- c) A right-angled triangle has shorter sides 18 cm and 32 cm. Find the exact hypotenuse in simplest surd form. (2 marks)

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Find largest square factor | 2 Split under the root: (square $\times$ rest) | 3 Use $(a \times b) = \sqrt{a} \times \sqrt{b}$ | 4 Square-root the square factor | 5 Answer: $k \sqrt{m}$ (m has no square factors)

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

Do I always find the LARGEST square factor first time?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 2 3 b) 3 2 c) 2 5 d) 5 2

Q2: a) 2 2 b) 4 2 c) 3 5 d) 5 3

Q3: a) 7 2 b) 10 2 c) 8 2 d) 10 3

Q4: a) 2 7 b) $3 \times 2 = 6$ 3 c) $2 \times 5 = 10$ 2 d) $5 \times 2 = 10$ 2

Q5: a) 4 3 + 2 3 = 6 3 b) 5 3 - 3 3 = 2 3 c) 6 2 + 3 2 = 9 2

Q6: a) $80 = (16 \times 5) = 4 \ 5$ b) 3 28 = 3 $(4 \times 7) = 3 \times 2 \ 7 = 6 \ 7$ c) $180 = (36 \times 5) = 6 \ 5$

Q7: a) $72 = 6 \ 2 \text{ cm}$ b) $4 \times 6 \ 2 = 24 \ 2 \text{ cm}$ c) Second square side = 3 2, first = 6 2. $6 \ 2 > 3 \ 2$, so first is larger.

Q8: a) $(a^2b) = (a^2) \times b = a \ b$ b) $(50x^4y^2) = (25 \times 2 \times x^4 \times y^2) = 5x^2y \ 2$ c) $\text{Hyp}^2 = 18 + 32 = 50$; hyp = $\sqrt{50} = 5 \ 2 \text{ cm}$
